D2 PROOF

Proof of the Height Bound of an AVL Tree

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1 Problem Statement

Taking the height of a single-node tree as 0, prove that an AVL tree of height h contains at least $F(h+3)-1$ nodes, where $F(k)$ denotes the kth Fibonacci number.

Hence, prove that:

$$\boxed{h < 1.45 \log_2(n+2)}$$

where n is the total number of nodes in the AVL tree.

2 Definition of an AVL Tree

An AVL tree is a self-balancing binary search tree in which the heights of the left and right subtrees of every node differ by at most 1.

Mathematically,

$$|h_L - h_R| \leq 1.$$

The height of a node is defined as:

$$h = 1 + \max(h_L, h_R).$$

We assume:

- Height of an empty tree $= -1$
- Height of a single-node tree $= 0$

3 Minimum Number of Nodes in an AVL Tree

Let $N(h)$ represent the minimum number of nodes required to construct an AVL tree of height h.

For an AVL tree of height h, at least one subtree must have height $h-1$.

Since the AVL balancing condition allows the heights of two subtrees to differ by at most 1, the other subtree must have height at least $h-2$.

To minimize the total number of nodes, we choose the subtree heights as $h-1$ and $h-2$.

Therefore,

$$N(h) = 1 + N(h-1) + N(h-2).$$

The additional 1 represents the root node.

Hence, the recurrence relation is:

$$\boxed{N(h) = N(h-1) + N(h-2) + 1}$$

Initial Conditions

For an empty tree:

$$N(-1) = 0.$$

For a single-node tree:

$$N(0) = 1.$$

Now,

$$N(1) = 1 + N(0) + N(-1) = 1 + 1 + 0 = 2.$$

Similarly,

$$\begin{aligned} N(2) &= 1 + N(1) + N(0) = 1 + 2 + 1 = 4, \\ N(3) &= 1 + N(2) + N(1) = 1 + 4 + 2 = 7, \\ N(4) &= 1 + N(3) + N(2) = 1 + 7 + 4 = 12. \end{aligned}$$

Thus, the values of $N(h)$ are:

Height h	$N(h)$	$N(h)+1$
0	1	2
1	2	3
2	4	5
3	7	8
4	12	13
5	20	21

We observe that the sequence $N(h) + 1$ is:

$$2,\ 3,\ 5,\ 8,\ 13,\ 21,\ \ldots$$

This is the Fibonacci sequence beginning from $F(3)$.

Therefore, we conjecture that:

$$\boxed{N(h) = F(h+3) - 1}$$

4 Proof Using Mathematical Induction

We prove that:

$$N(h) = F(h+3) - 1.$$

The Fibonacci sequence is defined as:

$$\begin{aligned} F(0) &= 0, \\ F(1) &= 1, \\ F(k) &= F(k-1) + F(k-2). \end{aligned}$$

Step 1: Base Cases

For $h = 0$:

$$\begin{aligned} N(0) &= 1, \\ F(3) - 1 &= 2 - 1 = 1. \end{aligned}$$

Therefore, the formula holds for $h = 0$.

For $h = 1$:

$$\begin{aligned} N(1) &= 2, \\ F(4) - 1 &= 3 - 1 = 2. \end{aligned}$$

Therefore, the formula holds for $h = 1$.

Step 2: Inductive Hypothesis

Assume that the formula holds for $h-1$ and $h-2$.

Therefore,

$$\begin{aligned} N(h-1) &= F(h+2) - 1, \\ N(h-2) &= F(h+1) - 1. \end{aligned}$$

Step 3: Inductive Step

Using the recurrence relation:

$$N(h) = 1 + N(h-1) + N(h-2).$$

Substituting the inductive hypotheses:

$$N(h) = 1 + \big[F(h+2) - 1\big] + \big[F(h+1) - 1\big].$$

Simplifying:

$$N(h) = F(h+2) + F(h+1) - 1.$$

Using the Fibonacci recurrence:

$$F(h+3) = F(h+2) + F(h+1).$$

Therefore,

$$\boxed{N(h) = F(h+3) - 1.}$$

Hence, by mathematical induction, the formula holds for all $h \geq 0$.

Since $N(h)$ is the minimum possible number of nodes in an AVL tree of height h, any AVL tree containing n nodes satisfies:

$$n \geq N(h).$$

Therefore,

$$\boxed{n \geq F(h+3) - 1.}$$

Hence, the first part is proved.

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5 Introduction to the Golden Ratio

To establish a logarithmic upper bound on the height, we need to express Fibonacci numbers in terms of an exponential function.

For this purpose, we introduce the golden ratio, denoted by φ.

The golden ratio is defined as:

$$\varphi = \frac{1+\sqrt{5}}{2}.$$

Numerically,

$$\varphi \approx 1.6180339887.$$

Derivation of Its Important Property

Consider the quadratic equation:

$$x^2 = x + 1.$$

Rearranging:

$$x^2 - x - 1 = 0.$$

Using the quadratic formula:

$$\begin{aligned} x &= \frac{1 \pm \sqrt{1+4}}{2} \\ &= \frac{1 \pm \sqrt{5}}{2}. \end{aligned}$$

Taking the positive root:

$$\varphi = \frac{1+\sqrt{5}}{2}.$$

Therefore,

$$\boxed{\varphi^2 = \varphi + 1.}$$

Multiplying both sides by φ^{k-2}:

$$\varphi^k = \varphi^{k-1} + \varphi^{k-2}.$$

This is similar to the Fibonacci recurrence:

$$F(k) = F(k-1) + F(k-2).$$

Thus, powers of the golden ratio satisfy the same type of recurrence as Fibonacci numbers. This property allows us to establish a lower bound on Fibonacci numbers.

6 Proof of the Fibonacci Lower Bound

We prove that:

$$\boxed{F(k) \geq \varphi^{k-2}}$$

for every integer $k \geq 2$.

We use mathematical induction.

Step 1: Base Cases

For $k = 2$:

$$F(2) = 1,$$
$$\varphi^{2-2} = \varphi^0 = 1.$$

Therefore,

$$F(2) \geq \varphi^0.$$

For $k = 3$:

$$F(3) = 2,$$
$$\varphi^{3-2} = \varphi \approx 1.618.$$

Since $2 > 1.618$,

$$F(3) > \varphi.$$

Thus, both base cases hold.

Step 2: Inductive Hypothesis

Assume that the inequality holds for $k-1$ and $k-2$.

Therefore,

$$F(k-1) \geq \varphi^{k-3},$$
$$F(k-2) \geq \varphi^{k-4}.$$

Step 3: Inductive Step

Using the Fibonacci recurrence:

$$F(k) = F(k-1) + F(k-2).$$

Substituting the inductive inequalities:

$$F(k) \geq \varphi^{k-3} + \varphi^{k-4}.$$

Taking φ^{k-4} as a common factor:

$$F(k) \geq \varphi^{k-4}(\varphi + 1).$$

Since

$$\varphi + 1 = \varphi^2,$$

therefore,

$$F(k) \geq \varphi^{k-4} \times \varphi^2.$$

Using the exponent rule

$$\varphi^a \times \varphi^b = \varphi^{a+b},$$

we obtain:

$$F(k) \geq \varphi^{k-4+2}.$$

Therefore,

$$\boxed{F(k) \geq \varphi^{k-2}.}$$

Hence, by mathematical induction, the inequality holds for every integer $k \geq 2$.

7 Derivation of the AVL Height Bound

From Section 4, we proved that:

$$n \geq F(h+3) - 1.$$

Adding 1 to both sides:

$$\boxed{n + 1 \geq F(h+3)} \tag{1}$$

From Section 6, we proved that:

$$F(k) \geq \varphi^{k-2}.$$

Substituting $k = h + 3$:

$$F(h+3) \geq \varphi^{(h+3)-2}.$$

Therefore,

$$\boxed{F(h+3) \geq \varphi^{h+1}} \tag{2}$$

Combining equations (1) and (2):

$$n + 1 \geq F(h+3) \geq \varphi^{h+1}.$$

Hence,

$$\boxed{n + 1 \geq \varphi^{h+1}} \tag{3}$$

Step 1: Taking Logarithms

Taking logarithms to base 2 on both sides:

$$\log_2(n+1) \geq \log_2\big(\varphi^{h+1}\big).$$

Using the logarithmic identity

$$\log(a^b) = b\log(a),$$

we obtain:

$$\log_2(n+1) \geq (h+1)\log_2(\varphi).$$

Step 2: Isolating h

Dividing both sides by $\log_2(\varphi)$, which is positive:

$$\frac{\log_2(n+1)}{\log_2(\varphi)} \geq h+1.$$

Therefore,

$$h+1 \leq \frac{\log_2(n+1)}{\log_2(\varphi)}.$$

Subtracting 1:

$$\boxed{h \leq \frac{\log_2(n+1)}{\log_2(\varphi)} - 1}$$

Step 3: Evaluating the Constant

We know:

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.6180339887.$$

Therefore,

$$\log_2(\varphi) \approx 0.6942419.$$

Hence,

$$\frac{1}{\log_2(\varphi)} \approx 1.4404201.$$

Substituting:

$$h \leq 1.4404201\ldots\log_2(n+1) - 1.$$

Since

$$1.4404201\ldots < 1.45$$

and $\log_2(n+1) > 0$ for $n \geq 1$,

$$h < 1.45\ \log_2(n+1) - 1.$$

Also, since

$$n+1 < n+2,$$

we have

$$\log_2(n+1) < \log_2(n+2).$$

Therefore,

$$1.45\ \log_2(n+1) - 1 < 1.45\ \log_2(n+2).$$

Combining the inequalities:

$$\boxed{h < 1.45\ \log_2(n+2)}$$

Hence, the second part is proved.

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8 Conclusion

We first derived the recurrence relation for the minimum number of nodes in an AVL tree:

$$\boxed{N(h) = 1 + N(h-1) + N(h-2)}$$

Using mathematical induction and the Fibonacci recurrence, we proved:

$$\boxed{N(h) = F(h+3) - 1}$$

Therefore, an AVL tree of height h contains at least:

$$\boxed{n \geq F(h+3) - 1}$$

Next, using the golden ratio

$$\varphi = \frac{1+\sqrt{5}}{2}$$

and the inequality

$$F(k) \geq \varphi^{k-2},$$

we obtained:

$$n + 1 \geq \varphi^{h+1}.$$

Taking logarithms and simplifying, we proved:

$$\boxed{h < 1.45\ \log_2(n+2)}$$

Consequently,

$$\boxed{h = O(\log n)}$$

Thus, the height of an AVL tree grows logarithmically with the number of nodes.

This guarantees that search, insertion, and deletion operations in an AVL tree can be performed in $O(\log n)$ time.

Hence, both required results are proved.